INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Selected Topics in Cryptography

Lab Session 03: Toy ECDH *March 3, 2026*

You can use C/C++, C#, Java or Python to develop your source code, but please use only one of this languages to develop all the exercises. You **MUST NOT USE any cryptographic library to implement ECDH**. Please use projective coordinates i.e $(x, y, z)$to represent each point of the elliptic curve.

1 Scalar multiplication kP

Develop the following exercises by your own. Please represent a point in an elliptic curve using three coordinates (x, y, z). The point at infinity $\mathcal{O} = (0, 1, 0)$. Any other point in the elliptic curve will be represented as $(x, y, 1)$, where $x, y \in \mathbb{Z}_p$. Use this representation to do the following exercises.

Use the implementation to add and double points in an elliptic curve that you did in a previous lab session

1. Design a function that implements **Algorithm 1** *Right-to-left binary method for point multiplication*

Algorithm 1: Right-to-left binary method for point multiplication(k,P)

Input : $k = (k_{t-1}, \ldots, k_1, k_0)_2$ binary representation of k; $P \in \mathbb{E}(GF(p))$
Output: kP

$Q = \infty$; `/* infinity point */`
for $i \leftarrow 0$ **to** $t - 1$ **do**
    if $k_i == 1$ **then**
        $Q \leftarrow Q + P$; `/* point addition */`
    end
    $P \leftarrow 2P$; `/* point doubling */`
end
return Q

2. Design a function that implements **Algorithm 2** *Left-to-Right binary method for point multiplication*
3. Test each function (exercise 1 and 2) for at least 5 different examples.

Algorithm 2: Left-to-right binary method for point multiplication(k,P)

Input : $k = (k_{t-1}, \ldots, k_1, k_0)_2$ binary representation of k; $P \in \mathbb{E}(GF(p))$

Output: kP

$Q = \infty$; `/* infinity point */`

for $i \leftarrow t-1$ **to** 0 **do**

  $Q \leftarrow 2Q$; `/* point doubling */`

  if $k_i == 1$ **then**

    $Q \leftarrow Q + P$; `/* point addition */`

  end

end

return Q

2 Simulation of ECDH

Simulate the ECDH with elliptic curves protocol in pairs. **Use the implementation to add and double points in an elliptic curve that you did in a previous lab session**. Also use the functions of Section 1 to compute kP. To show the simulation every student must use her/his own computer and proceed as follows.

1. Do agree with your partner a prime number p such that $|p| \geq 7$ bits, the parameters of a non-singular elliptic curve a and b a generator point $G \in \mathbb{E}(a, b)$. Please consider that your program must accept large prime numbers, i.e. $|p| = 128, 256, 512, 1024$ bits.
2. One student will play the role of Alice and the other one will play the role of Bob.
3. Alice randomly choose an integer k_A and compute $A = k_A G$. Share A with Bob.
4. Bob randomly choose exponent k_B and compute $B = k_B G$. Share B with Bob.
5. Once Alice received the value B from the other student, must calculate $k_A B$
6. Once Bob received the value A from the other student, must calculate $k_B A$
7. Compare your results. A and B must be equal.

3 Products

Every student can write her/his own report including the following information:

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. The source code you did to implement algorithm 1 and algorithm 2 (Section 1)
3. Test your functions for Algorithm 1 and 2 as follows. Use at least 5 different examples. You must show the elliptic curve you used, the point P, the value of k and the output kP. Show the result of each iteration of the for cycle.